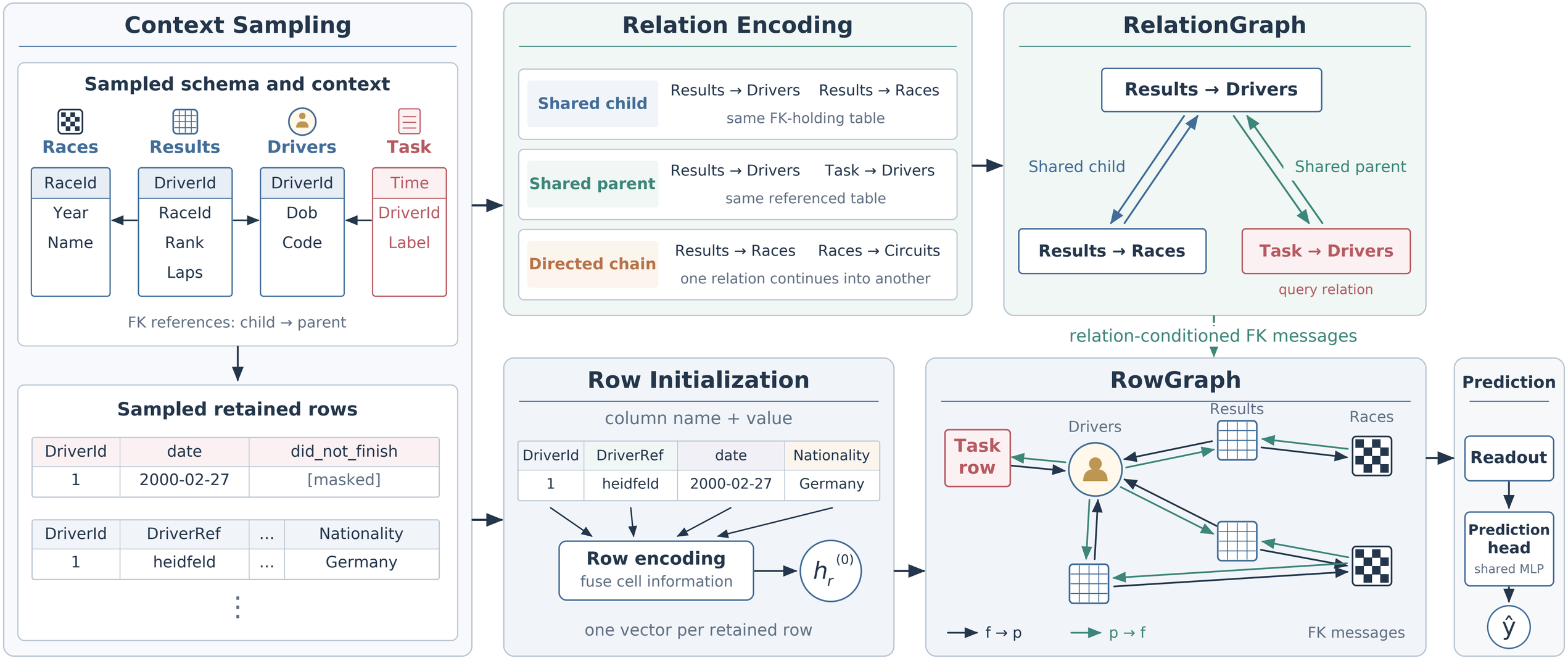
Context Sampling
Sampled schema and context
Races
Results
Drivers
Task
RaceId
Year
Name
DriverId
RaceId
Rank
Laps
DriverId
Dob
Code
Time
DriverId
Label
FK references: child → parent
Sampled retained rows
DriverId
date
did_not_finish
1
2000-02-27
[masked]
DriverId
DriverRef
...
Nationality
1
heidfeld
...
Germany
Relation Encoding
Shared child
Results → Drivers
Results → Races
same FK-holding table
Shared parent
Results → Drivers
Task → Drivers
same referenced table
Directed chain
Results → Races
Races → Circuits
one relation continues into another
RelationGraph
Results → Drivers
Shared child
Shared parent
Results → Races
Task → Drivers
query relation
relation-conditioned FK messages
Row Initialization
column name + value
DriverId
DriverRef
date
Nationality
1
heidfeld
2000-02-27
Germany
Row encoding
fuse cell information
$h_r^{(0)}$
one vector per retained row
RowGraph
Task row
Drivers
Results
Races
f → p
p → f
FK messages
Prediction
Readout
Prediction head
shared MLP
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